

Clustering II

Rachael Hageman Blair

Outline

- Intro to K-means algorithm
- Minimization Criterion
- Limitations
- Soft k-means
- Example
- Extensions

Kmeans: Algorithm

Algorithm 14.1 *K -means Clustering.*

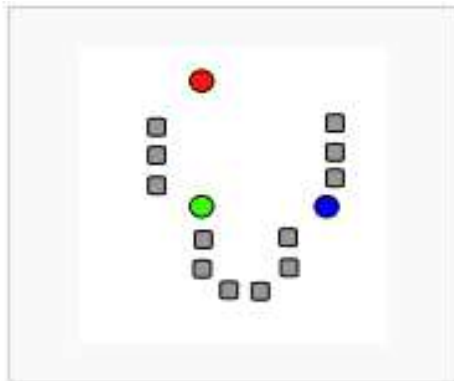
1. For a given cluster assignment C , the total cluster variance (14.33) is minimized with respect to $\{m_1, \dots, m_K\}$ yielding the means of the currently assigned clusters (14.32).
2. Given a current set of means $\{m_1, \dots, m_K\}$, (14.33) is minimized by assigning each observation to the closest (current) cluster mean. That is,

$$C(i) = \underset{1 \leq k \leq K}{\operatorname{argmin}} \|x_i - m_k\|^2. \quad (14.34)$$

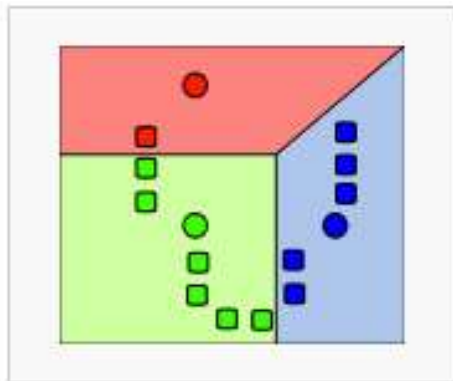
3. Steps 1 and 2 are iterated until the assignments do not change.
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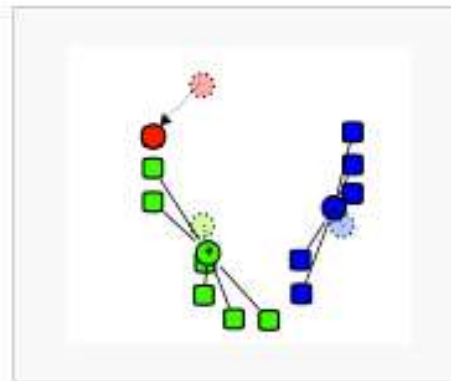
Kmeans: Algorithm



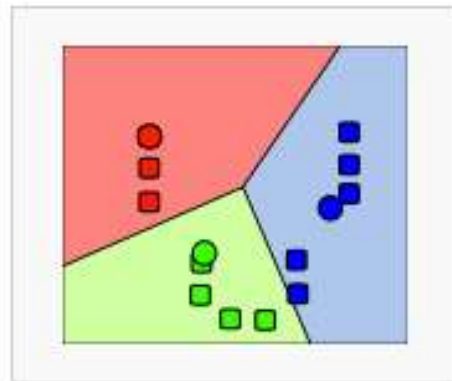
1) k initial "means" (in this case $k=3$) are randomly generated within the data domain (shown in color).



2) k clusters are created by associating every observation with the nearest mean. The partitions here represent the [Voronoi diagram](#) generated by the means.



3) The [centroid](#) of each of the k clusters becomes the new mean.



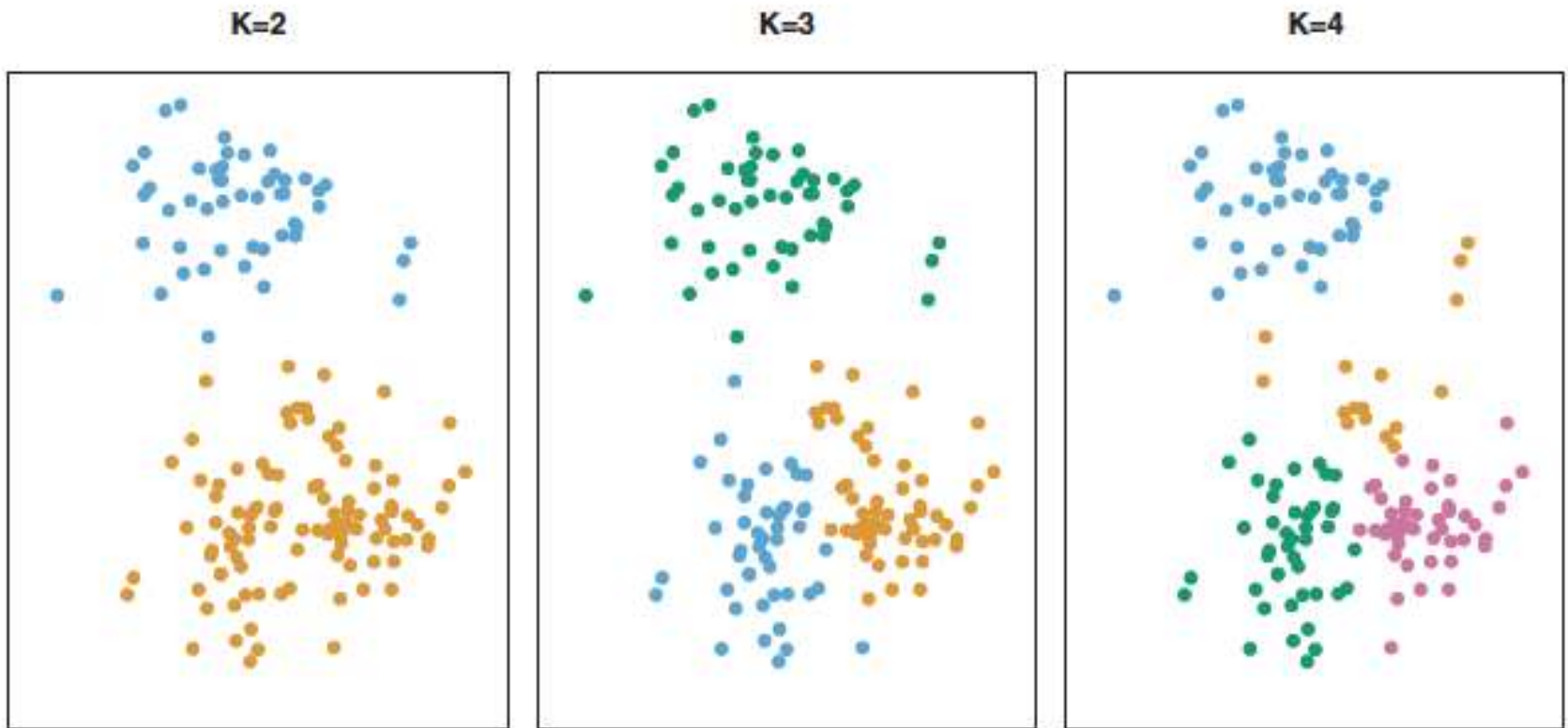
4) Steps 2 and 3 are repeated until convergence has been reached.

Kmeans: Algorithm

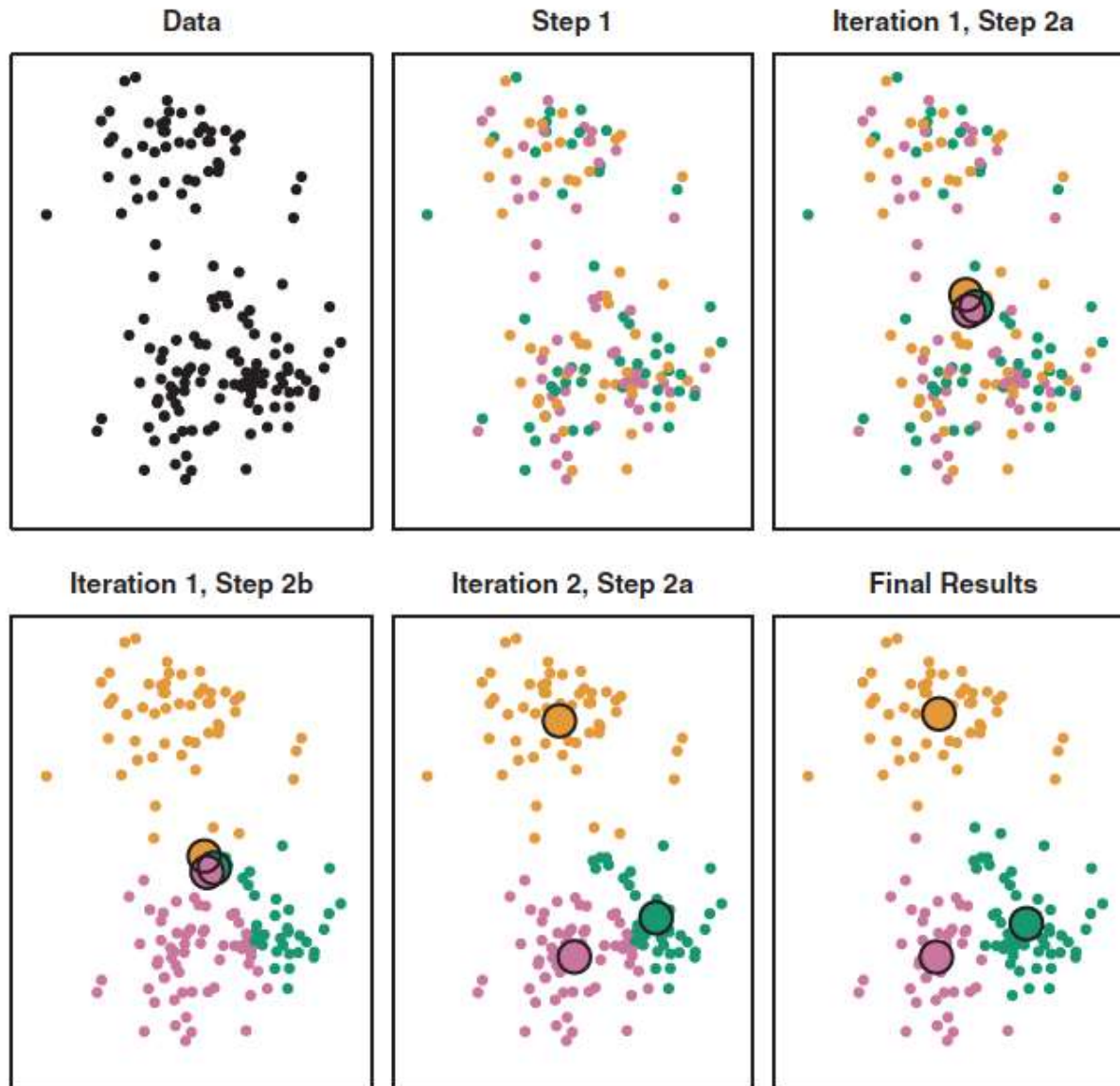
Computational Issues:

- May get stuck at local minima
- Include a number of different starting points
- Can include constraints, e.g, no single switches of one observation from one group to another group, which will decrease the objective function.

Kmeans: Algorithm



Kmeans: Algorithm



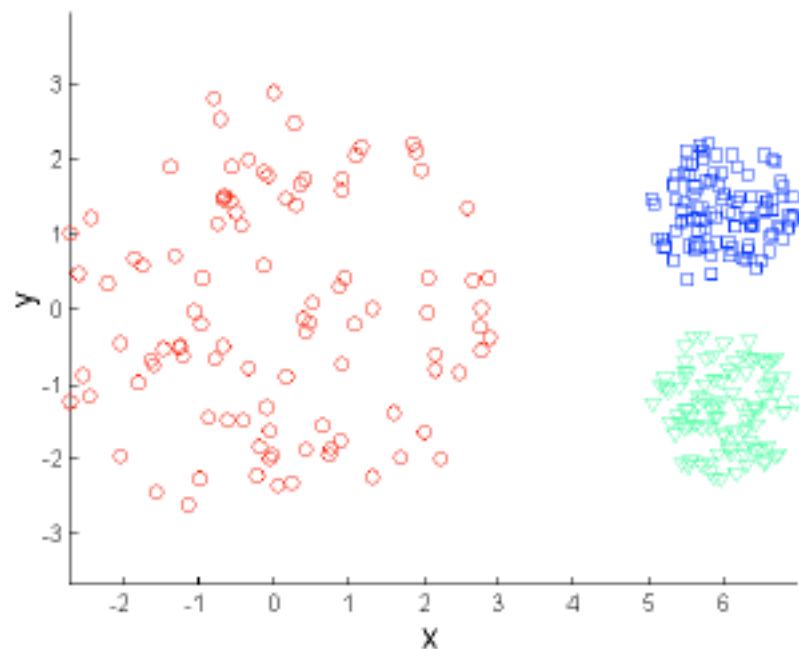
Kmeans: Limitations

- Sensitive to outliers
- Empty clusters can appear

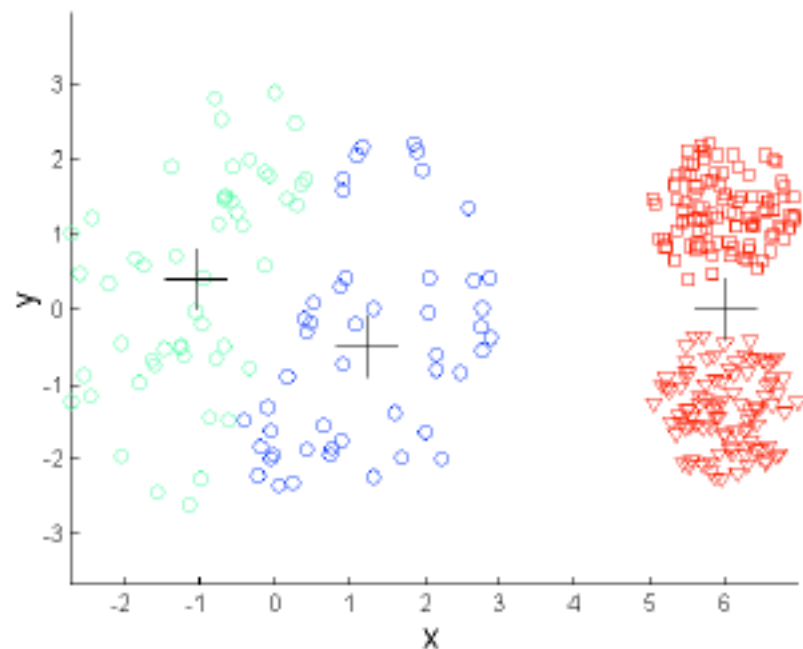
K means may be unreliable when clusters differ in:

- Size
- Density
- Odd shape

Kmeans: Limitations

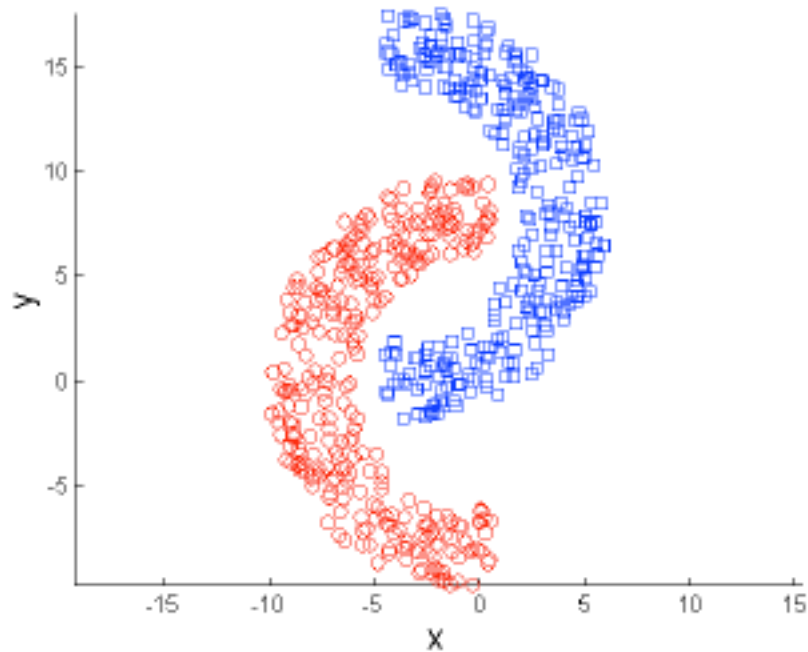


Original Points

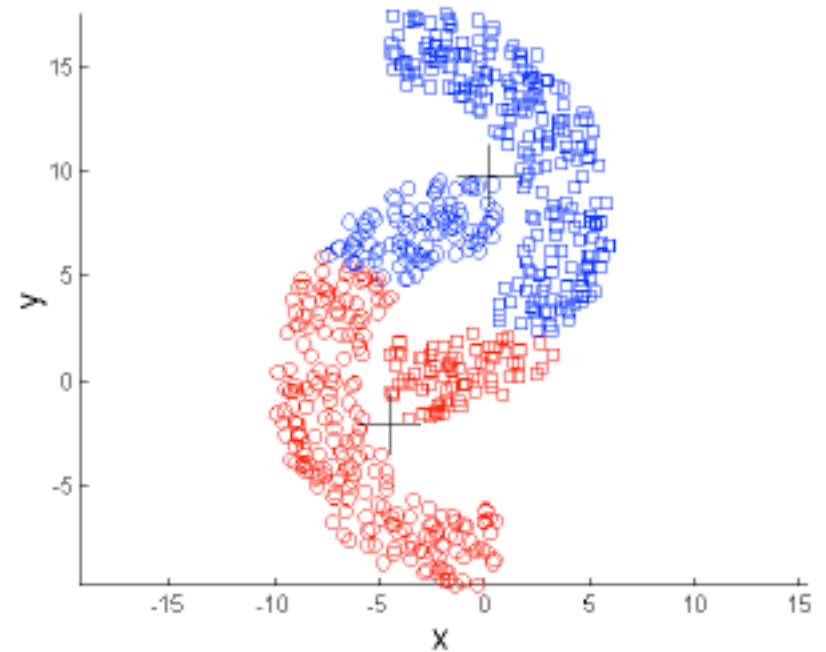


K-means (3 Clusters)

Kmeans: Limitations

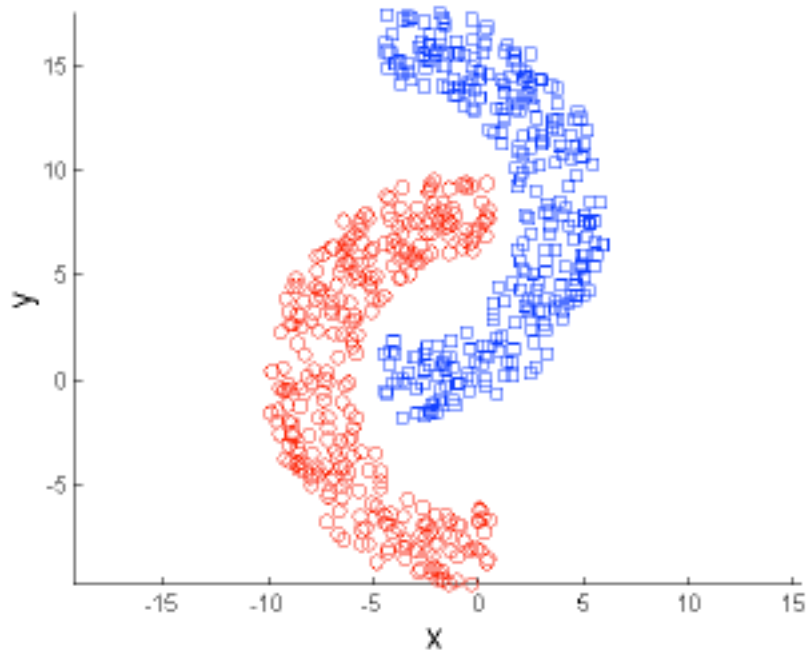


Original Points

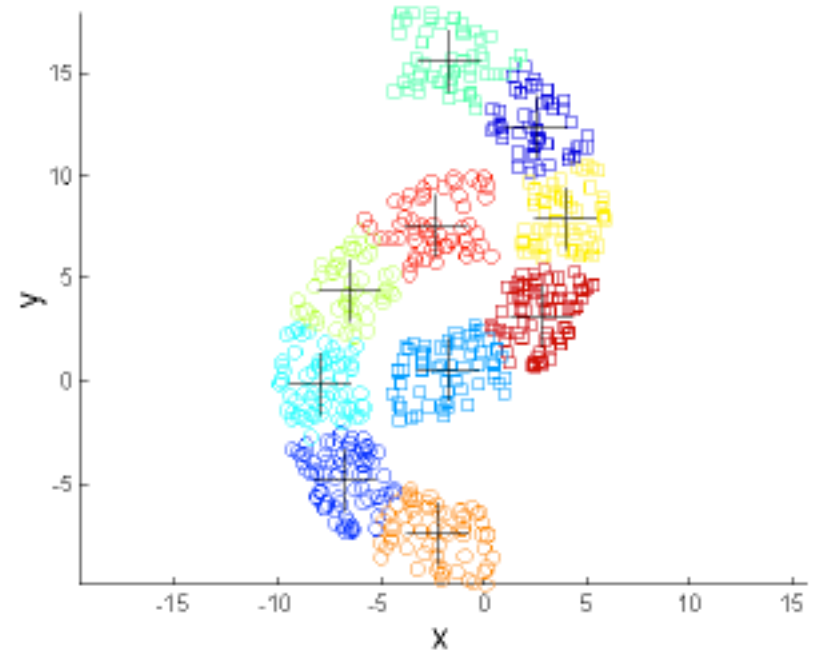


K-means (2 Clusters)

Kmeans: Limitations



Original Points



K-means Clusters

Soft k-means

- **Goal:** minimize the sum of square distances of each data point to the center of its group.
- **Before:** A data point is a member of a group, or it is not (binary).
- Soft k-means, recognizes that a data point may belong to more than one group with a certain probability (uncertainty).

Gaussian Mixtures as “soft k-means”

- K-means is closely related to the Expectation Maximization (EM) algorithm for estimating a Gaussian mixture model.
- A “soft” version of k-means by shifting from deterministic to probabilistic assignments of points to clusters.

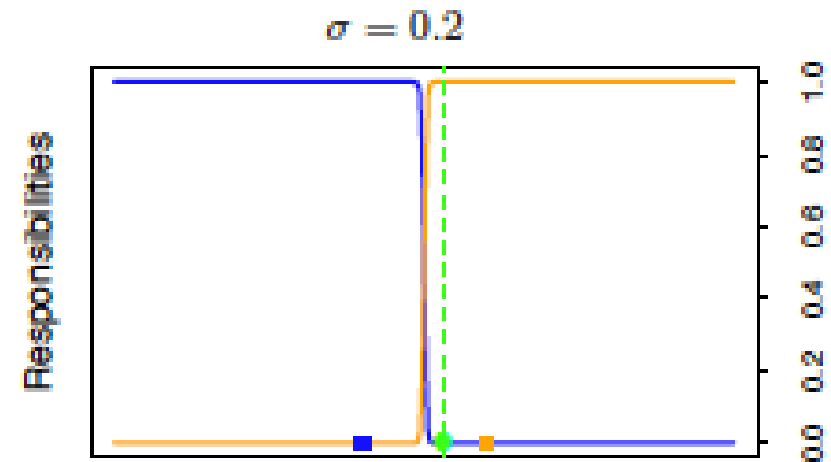
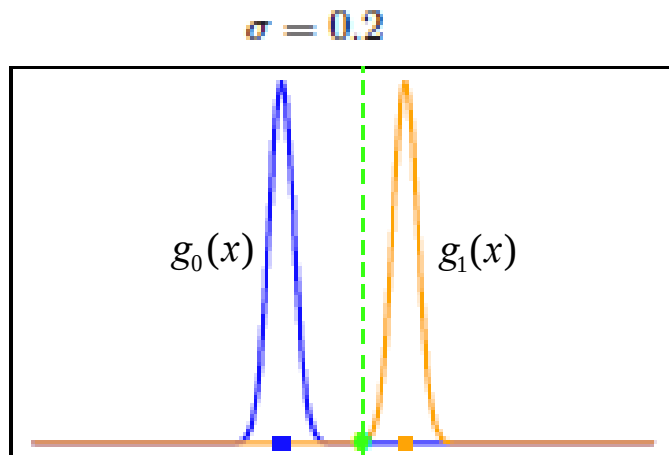
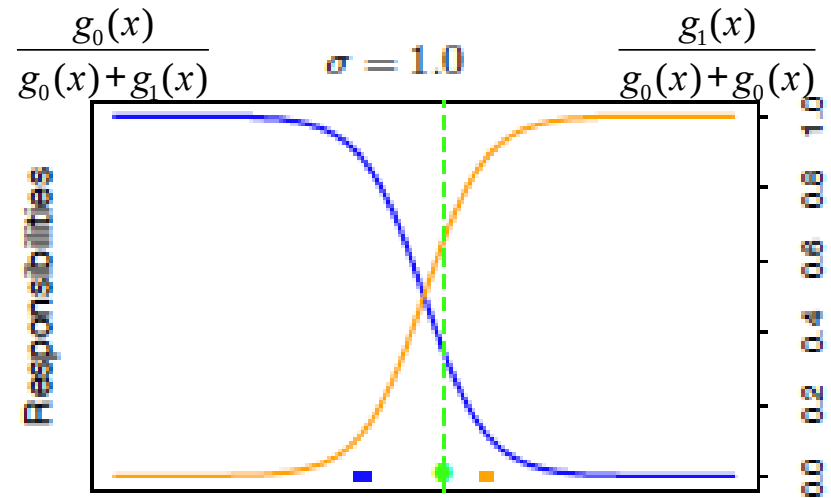
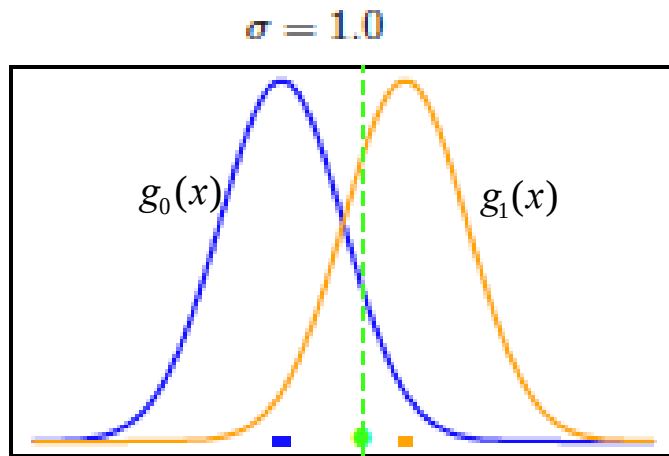
In this setting:

1. **E-step:** assigns “responsibilities” for each data point based on its relative density under each mixture component.
2. **M-step:** recalculate the component density parameters based on the current “responsibilities”.

Gaussian Mixtures as “soft k-means”

- Suppose K mixture components are specified, each with a Gaussian density with a scalar covariance matrix $\sigma^2 I$.
- The relative density under each mixture component is a monotone function of the Euclidean distance between the data point and the mixture center.
- As $\sigma^2 \rightarrow 0$, these probabilities become 0 and 1, and the two methods coincide.

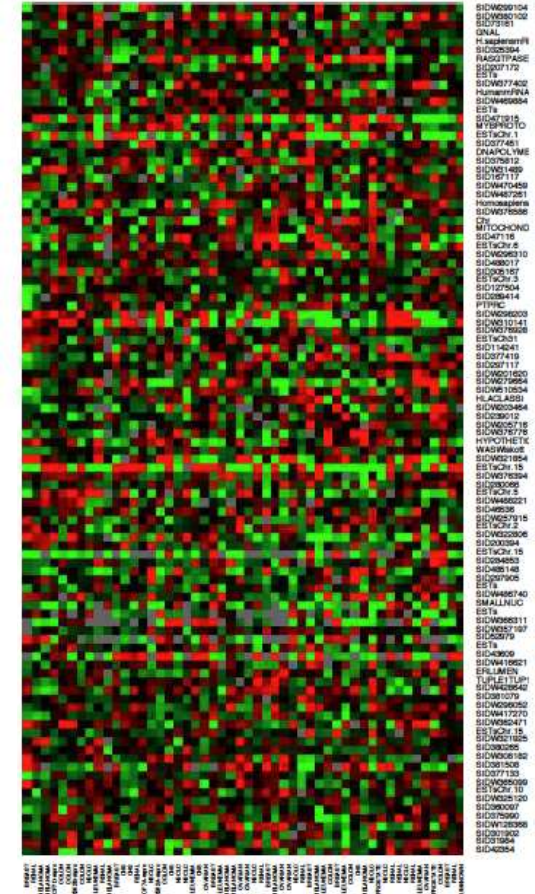
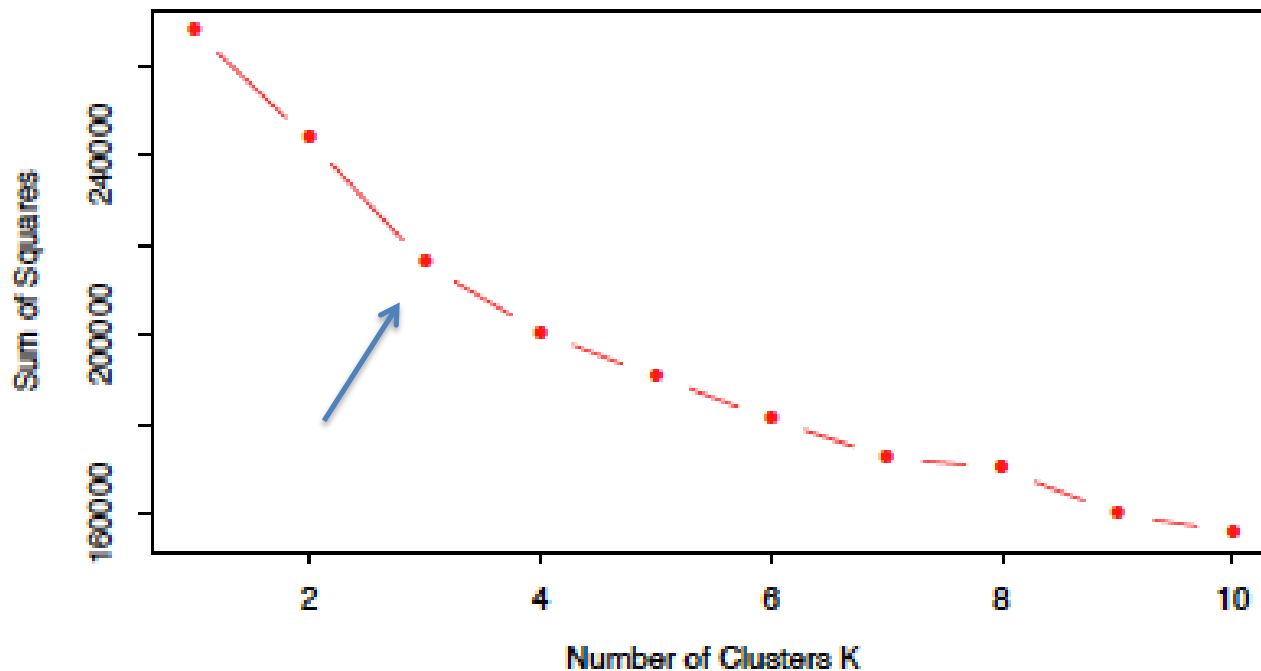
Soft K-means



An Example

Human Tumor Microarray

- 6830 transcripts (genes) and 64 samples



An Example

TABLE 14.2. *Human tumor data: number of cancer cases of each type, in each of the three clusters from K-means clustering.*

Cluster	Breast	CNS	Colon	K562	Leukemia	MCF7
1	3	5	0	0	0	0
2	2	0	0	2	6	2
3	2	0	7	0	0	0
Cluster	Melanoma	NSCLC	Ovarian	Prostate	Renal	Unknown
1	1	7	6	2	9	1
2	7	2	0	0	0	0
3	0	0	0	0	0	0

K-means further reading



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Award winning papers from the 19th International Conference on Pattern Recognition
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19th International Conference in Pattern Recognition (ICPR)



Data clustering: 50 years beyond K-means ☆

Anil K. Jain 🧑 📧

Department of Computer Science and Engineering, Michigan State University, East Lansing, Michigan 48824, USA

Department of Brain and Cognitive Engineering, Korea University, Anam-dong, Seoul, 136-713, Korea

✉️ anil.k.jain@msu.edu | www.cse.msu.edu/~anil | www.bce.ku.ac.kr/~anil